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Linear Regression for Humans: Predicting the Future in Plain English

A friendly guide to understanding linear regression and how to actually explain it

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 Data Science Collective a11y-light ~9 min read · August 31, 2025 (Updated: August 31, 2025) ·

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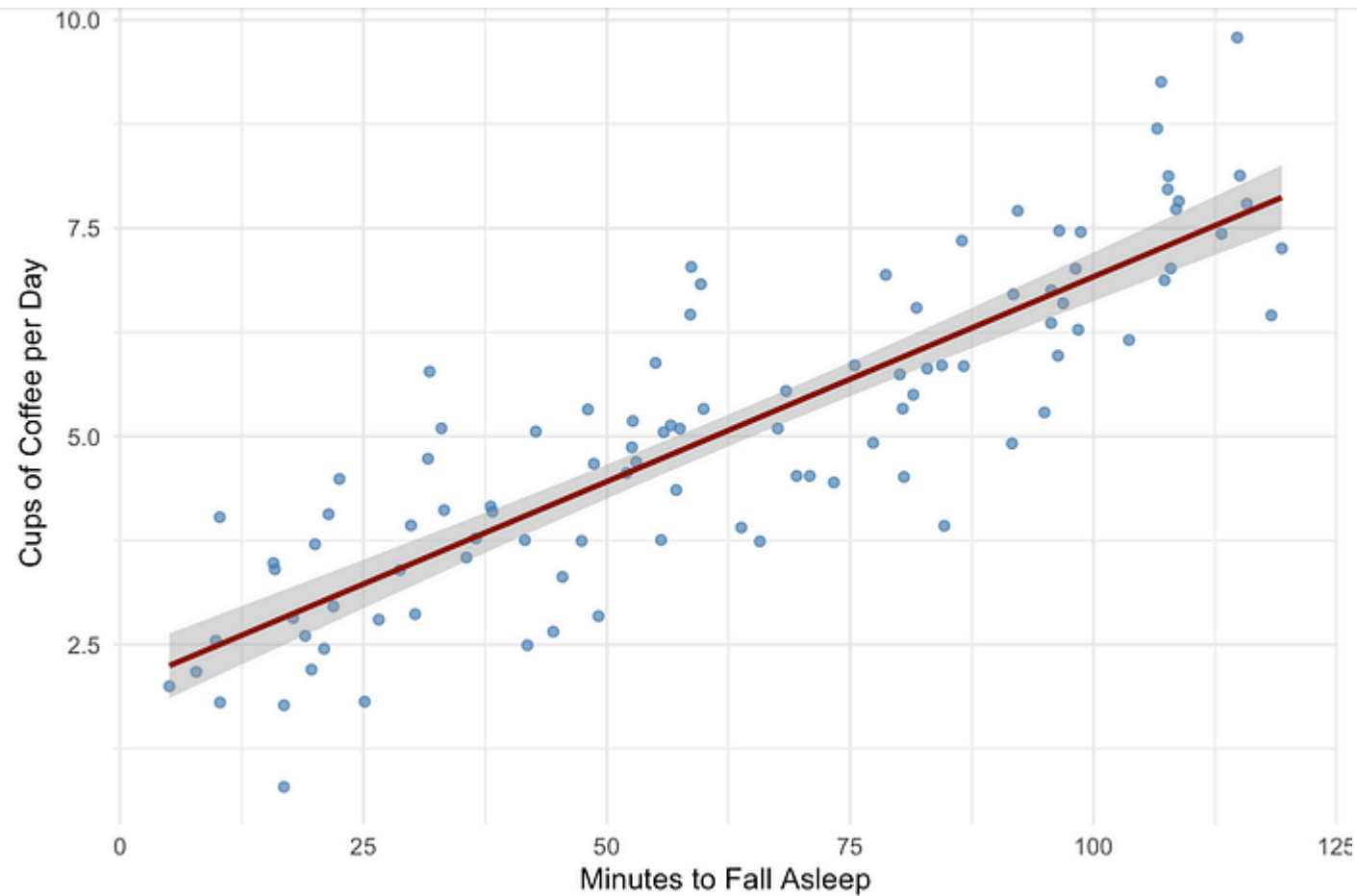
You probably already do linear regression in your head:

- *The more coffee I drink, the harder it is to fall asleep.*
- *The more it rains, the greener the grass looks.*
- *The older I get, the worse my knees feel.*

Linear regression is the science of drawing the best straight line through your data to help you understand relationships and make predictions. It's the tool for finding that overall relationship even when individual points don't follow it exactly.

There are outliers to these situations of course. Not everyone has problems with their knees, and caffeine doesn't affect every person equally.

But if you surveyed 500 people and asked them how much coffee they drink and how long it took for them to fall asleep, you would get a scatter plot that is trending upwards. You could draw a straight line right through the middle, and it might look something like this:



R plot by author

Real life is messy. Some tall people weigh less than some short people. Some study less but get higher grades.

But if you take a step back, you might see a trend:

• MORE studying tends to lead to higher scores.

In this article, we'll explore linear regression in plain English without code or advanced math. Just pizza-sized examples that you can actually understand.

How Is The Line Drawn?

Here's the important part from statistics class: the "best fit line" is chosen by minimizing errors. This method is called ordinary least squares (OLS). Think of it as the line that's *least* wrong overall. In other words, the line is picked so the distance between the actual dots and the line are as small as possible on average.

Imagine every point in the graph above is tied to the line with a tiny rubber band. If the line is far away, the band stretches tight. OLS is like adjusting the line until the total stretch across all bands is as small as possible.

A Simple Regression Example: Pizza Prices and Size

You are browsing the menu at your favorite pizza restaurant. You notice that bigger pizzas have higher prices. So you gather the data.

Size (inches)	Price (\$)
10	8
12	10
14	12
16	14

Image by author

You can get graph paper and plot these points like you did in algebra class, and it would confirm that there is a relationship between pizza size and cost.

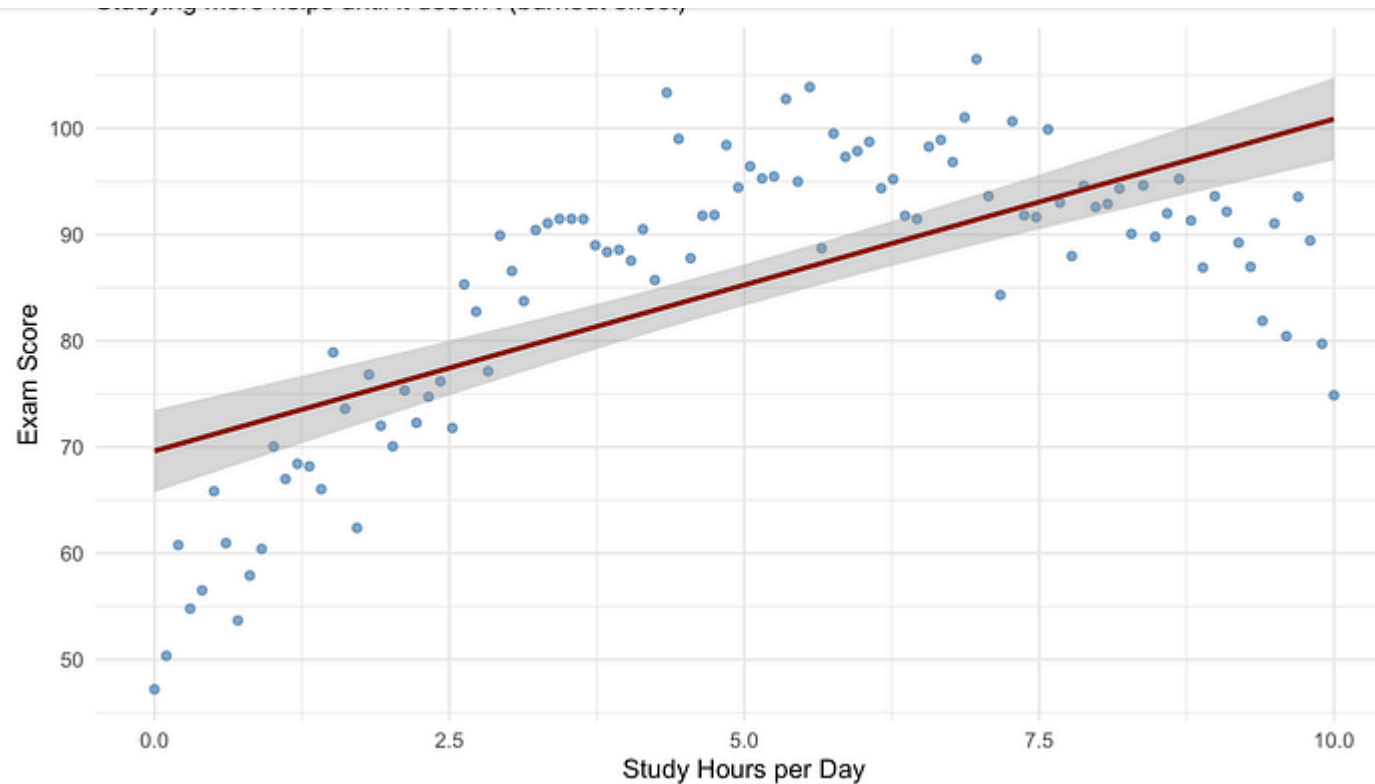
Linear regression is the math behind that line.

So if someone asks: "How much would a 15-inch pizza cost?", you could look at the line and say, "Probably around \$13." And be careful, it's still a prediction, not an exact number. Predictions can be wrong.

- In **statistics**, linear regression is often taught as a way to explain relationships ("how much does x change when y changes?").

In reality, you wouldn't build a machine learning model for pizza price, but both use the same underlying math: finding the line that minimizes error.

But here's the thing, we can't assume every relationship is linear. If the relationship is curved (e.g., age vs. salary often rises then flattens), a straight line won't capture it, and a linear regression model won't work. Take the example below: the line doesn't fit.



R plot by author

What Does Linear Regression Tell Us?

So, if you work in any field where people ask *how much*: how much will sales increase or how much will a customer spend, linear regression is usually the first method you try as a data scientist. It predicts a **continuous** outcome.

"We used past data to draw a line. That line helps us guess the future."

Besides making predictions, linear regression can tell you:

- **Direction** of the relationship: Does more of x (size) lead to more of y (price)?
- **Strength** of the relationship: Is the line tight or are the dots spread out everywhere?
- **Importance**: Which variable helps predict the outcome best?

A linear regression model will also give you some more technical outputs:

- **R^2 (R-squared)**: How much of the variation in what you're predicting is explained by the line. Higher R^2 = tighter fit.
- **p-values**: Whether a variable is statistically useful or just noise.
- **Coefficients (β)**: The weights applied to each variable that show how much they matter in the prediction.

But we won't get too much in the weeds on model metrics in this post. Next, let's look at the linear regression equation. Don't worry, I'll explain each part.

Linear Regression Ingredients

Here's the classic statistics linear regression equation:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \varepsilon$$

This might look a little scarier than the **slope-intercept form** equation from algebra class $y = mx + b$, but it's basically saying the same thing, just with proper statistical notation.

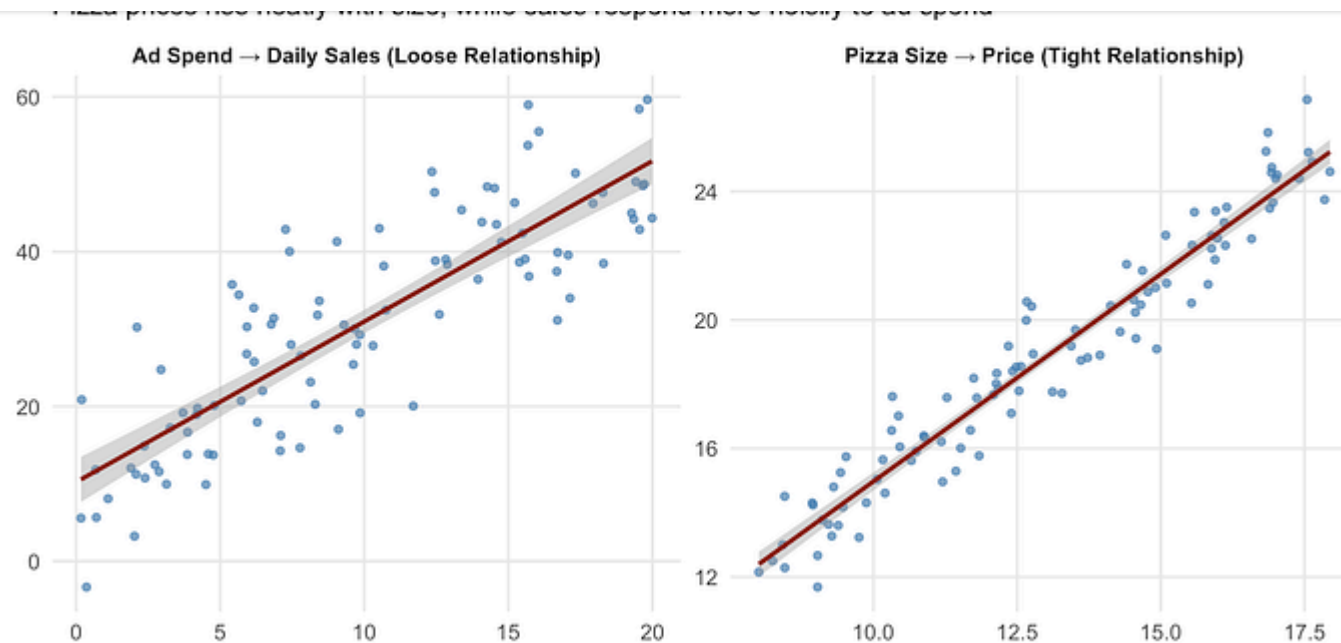
You don't have to memorize any equations or know how to calculate anything by hand, but it helps to know what goes on behind the scenes.

So, let's break it down:

- $\hat{y}$ is your **predicted value** — what you're trying to estimate (like the price of a pizza). We call it a **dependent variable** or "y hat".
- x is your **input** — the thing you're using to make that prediction (the size of the pizza). We can use more than one variable to predict $\hat{y}$, in which case we add a $\beta_2 x_2$ to the equation, and so on. In the pizza example, you could also add premium toppings, cheese level, and crust type as input variables to

- β_1 is the **slope** — how much $\hat{y}$ changes for each one-unit increase in x (e.g., how much price increases for every extra inch). We also call these **weights** or **coefficients** applied to each x variable.
- β_0 is the **intercept** — the predicted value when $x = 0$ (often not meaningful, but it's part of the math). We also call this "beta naught". It sets the starting point of the line before the other variables (slopes) adjust it up or down.
- ϵ is the **error term** — the "noise" or randomness we can't explain. In machine learning, we like to assume this gets averaged out, but it's important in statistics.

A linear relationship can be loose (lots of noise) or tight (less noise). Consider the example below:



R plot by author

Be careful though, just because two things move together doesn't mean one causes the other. More firefighters at a fire doesn't cause more damage; the fire size drives both.

We also have to be careful not to add too many informative features to our model, which will make it messy and unable to make good predictions on unseen data. Variables that measure almost the same thing (like "pizza size" and "number of slices") can confuse the model. This is called **multicollinearity**.

sometimes it's not.

And if you decide you want to add more data, you can go long (find more rows or samples) or go wide (add more features).

- **Deep** (more rows/observations): better for making your model more reliable and less noisy.
- **Wide** (more columns/features): better for making your model more explanatory or capturing more complexity.

Creating more features with existing data is called **feature enrichment** or **feature engineering**. For example, if your dataset has pizza size and number of slices, you might create a new feature like price per slice. That's feature engineering.

But too much of this and our model can start memorizing quirks in the training data instead of learning a general pattern. Creating new features doesn't fix low-quality data. If the raw inputs are noisy or biased, engineered features just amplify the noise. (Garbage in, garbage out.)

Classification vs. Linear Regression: What's the Difference?

"Can you tell me if a customer will order pepperoni or cheese?"

That's a **classification** problem.

You're assigning a **label**, in this case, "pepperoni" or "cheese." You're predicting categories, like:

- Spam or Not Spam
- High Risk or Low Risk

The model's job is to **predict a class** based on certain x variables which could be day of the week, time of day, age, location, etc. The goal could be a binary classification (two options) or multi-class (many options). The key is that you're predicting categories, not continuous numbers on a scale.

Later that day, your other manager asks:

"Can you estimate how much the customer will spend on their order?"

You're predicting a **number** — like \$18.75.

Regression models are used when your outcome is **continuous** (it can take on a range of values), such as:

- How far?
- How long?
- What score?

Like classification, the linear regression prediction is also based on certain x variables which could be day of the week, time of day, drink order, delivery or pickup, location, etc.

Classification predicts *what kind*. Regression predicts *how much*.

How Is This Different Than Time Series?

If you've read about time series, you know one thing is sacred: the order of the data.

But with linear regression, it's the opposite:

- Each row should be **independent**.
- Random train/test splits are okay, as long as the future doesn't leak into the past.
- Autocorrelation (points depending on one another) violates regression assumptions.

So while time series models predict what happens next, regression models answer what matters most right now.

Explaining vs. Predicting

Now that you know what linear regression is, I want to mention a nuance that's easy to miss:

Regression has two identities. It can be explanatory, or it can be predictive, but not always both at the same time.

understanding relationships.

- **Predictive regression** might say: "Based on size, features, and location, this house should sell for \$350,000." That's focused on getting the number right, not telling a story.

There are business use cases for both of these methods. You can build a linear regression model that either explains or predicts.

Both use the same math, but the focus is different: one is about understanding relationships, the other about making accurate guesses.

Takeaways

From estimating pizza prices to forecasting revenue, linear regression is one of the most widely used tools in data science. It's often the first model you reach for when someone asks "how much?"

Here are some important points to keep in mind about explaining linear regression:

- It predicts continuous numbers (like price or revenue), not categories.
- Coefficients (β) show how much each variable contributes to the prediction.
- R^2 tells you how well the line fits, while p-values help you spot which features actually matter.
- Correlation is not causation.
- Outliers and multicollinearity can distort your model, so data cleaning and feature selection are important.
- Adding more features (feature engineering) can help, but too many can lead to overfitting.
- Regression can be explanatory ("every year of experience adds \$5k to salary") or predictive ("this house should sell for \$350k"), but rarely both at once.

Linear regression won't give you perfect answers, but it will give you a clear, interpretable starting point. Hopefully now you'll be more confident the next time someone asks, "How much will it cost?"

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